

Useful Python Standard Library Functions for Working with Files

Let's take a look at some functions that could help you as you write your own match functions. The standard library of modules that comes with Python features several useful functions for getting information about files. Many of these are in the `os` and `shutil` modules, so your program must run `import os` OR `import shutil` before it can call these functions.

Finding Information About the File's Name

The full filepath passed to the match functions can be broken into the base name and directory name with the `os.path.basename()` and `os.path.dirname()` functions. You can also call `os.path.split()` to obtain these names as a tuple. Enter the following into Python's interactive shell. On macOS or Linux, try using `/bin/ls` as the filename:

Python

```
>>> import os
>>> filename = 'C:/Windows/system32/notepad.exe'
>>> os.path.basename(filename)
'notepad.exe'
>>> os.path.dirname(filename)
'C:/Windows/system32'
>>> os.path.split(filename)
('C:/Windows/system32', 'notepad.exe')
>>> folder, file = os.path.split(filename)
>>> folder
'C:/Windows/system32'
>>> file
'notepad.exe'
```

You can use any of Python's string methods on these string values to help evaluate the file against your search criteria, such as `lower()` in the `hasEveryVowel()` match function.